Ajay Venkatesh

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Education

New York University | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** Aug 2024 – May 2026

Skills

Programming Languages: Python, Java, C++, JavaScript (ES6), TypeScript, SQL
AI/ML: PyTorch, TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Hugging Face, SVM, Random Forest, MLflow, PySpark
Web Development: REST API, GraphQL API, Microservices, React, Node.js, Express.js
Database: RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse
Cloud: AWS (AWS CDK, API Gateway, Lambda, IAM, Q, SageMaker), Azure (AI services, Function Apps, Storage blobs)
Tools & Platforms: Linux, Git, Docker, Agile/Scrum, GitLab CI, Terraform, Ansible, Jupyter, Postman, Visual Studio, Boto3, Datadog

Experience

Campus Mesh — Jan 2026 – Present

AI Software Engineer — *New York, US*

- Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Designed and implemented a collaborative-filtering **recommendation engine** using user interaction signals, increasing platform engagement by **28%** and user retention by **50%**.
- Authored **LLM-based retrieval pipelines** for CampusIQ (AI Assistant), improving contextual accuracy by **30%** through semantic retrieval and structured data indexing.

Amazon — May 2025 – Aug 2025

Software Development Engineer Intern — *Seattle, US*

- Built a high-throughput, event-driven incident triage system using **AWS ECS Fargate Tasks, SQS, and SNS**, decoupling message processing and reducing incident response latency by **50%** during peak traffic.
- Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an agentic workflow that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
- Improved large-scale log analysis by implementing a **Model Context Protocol (MCP)** server for efficient retrieval, cutting AWS Athena query time and compute costs by **40%**.
- Provisioned resilient **Infrastructure-as-Code (IaC)** using **AWS CDK** to manage auto-scaling ECS clusters, ensuring **99.9%** availability under 2x projected load.

New York University (Novel AI Technologies, Inc.) — Aug 2024 – Present

AI Software Engineer — *New York, US*

- Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented time-series anomaly detection using **Isolation Forest** and **LSTM** models.
- Built **computer vision pipelines (YOLOv8, EfficientNet)** to process user-submitted food images integrated with **GPT-4 workflows** for nutritional estimation and dietary feedback.
- Developed an insurance analytics platform **(React, Node.js, REST APIs)** integrating **LLM-based document analysis (GPT-4 + LangChain)** for underwriting insights.
- Engineered ETL pipelines transforming **NoSQL** data into **PostgreSQL** with concurrency control; processed **250K+** records and powered **AWS QuickSight** dashboards.

PricewaterhouseCoopers (PwC) — Aug 2021 – Aug 2024

Senior Software Engineer

- Designed and implemented a distributed backend system on **Microsoft Azure** integrating with **Microsoft Dynamics** and supporting ML inference pipelines for healthcare analytics.
- Led code reviews and mentored interns on **software design and test-driven development**, reducing production bugs by **25%**.
- Built an **NLP-powered OCR pipeline** using **LayoutLM** and **Azure Form Recognizer** to automate data extraction from medical documents, reducing processing time by **75%**.
- Optimized data ingestion using **parallel processing** for **100K+** Excel records, reducing execution time from **5 hours to 1 hour**.

Projects

RV32I Processor Simulator (Single-Stage & 5-Stage Pipelined Core) | ***Python, RISC-V ISA, Computer Architecture***

- Built a cycle-accurate **RV32I simulator** in **Python** with single-stage and **5-stage pipelined (IF/ID/EX/MEM/WB)** cores, implementing R-type, I-type, LW/SW, BEQ/BNE, JAL with opcode-based dispatch.
- Engineered **hazard detection** with **EX/MEM-to-EX forwarding** and load-use stalls, resolving RAW dependencies without state rollback.
- Generated per-cycle register/memory/pipeline dumps and computed **CPI & IPC** metrics, validated across multiple assembly test cases.

InviLite (AI-Driven Data Engineering & Analytics Platform) | ***AWS, SageMaker, Bedrock, FAISS, Python***

- Architected a cloud-native data infrastructure (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
- Built a **RAG system** (SVM classification + FAISS vector indexing + Bedrock LLM), reducing incident resolution from **1–2 hours to minutes** and improving historical match accuracy by **30%**.
- Increased root-cause classification accuracy from **56% to 92%** through structured feature engineering and model retraining.